

HES Architecture and Operations

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Knowledge Base: Smart Meter AI Assistant — Synthetic Business Document

HES overview

The Head End System communicates with smart meters and manages sessions, commands, responses, events, and configured data collection.

Communication

HES communication involves network connectivity, protocol handling, command transmission, meter response, and acknowledgement.

Command processing

Commands should use approved workflows. Operations should monitor status, response time, retries, failures, and error information.

RC/DC

Remote Connect and Remote Disconnect are controlled commands. Their lifecycle should be monitored from initiation to final outcome.

Data collection

HES can collect reads, interval data, events, alarms, and other configured datasets. Collection schedules should be monitored for gaps.

Events

Events may include power failure, restoration, tamper, voltage abnormalities, and communication conditions.

Monitoring

Operational dashboards should track communication success, command success, latency, repeated retries, and failed devices.

Operational note

This is synthetic content created for local RAG development, testing, interview demonstration, and GitHub publication. It is not a confidential utility procedure or contractual document.